



Figure 1: OncoReconcile AI logo

OncoReconcile AI

AI-Powered Precision Oncology Data Quality, Semantic Harmonization, Governance, Interoperability & Analytics

DFWIT 2026 AI & Startup Competition — Final Submission

Team Variant Vanguard

Final branch:

<https://github.com/oncoreconcile-ai/oncoreconcile-ai/tree/enterprise-patient-journey-demo>

Demo video:

[FINAL DEMO VIDEO LINK — TO BE ADDED]

Submission date: July 11, 2026

The Opportunity

Precision oncology depends on data from electronic health records, molecular laboratories, research systems, clinical trials, and external partners. The same disease, gene, variant, treatment, or laboratory concept is frequently represented with different names and codes.

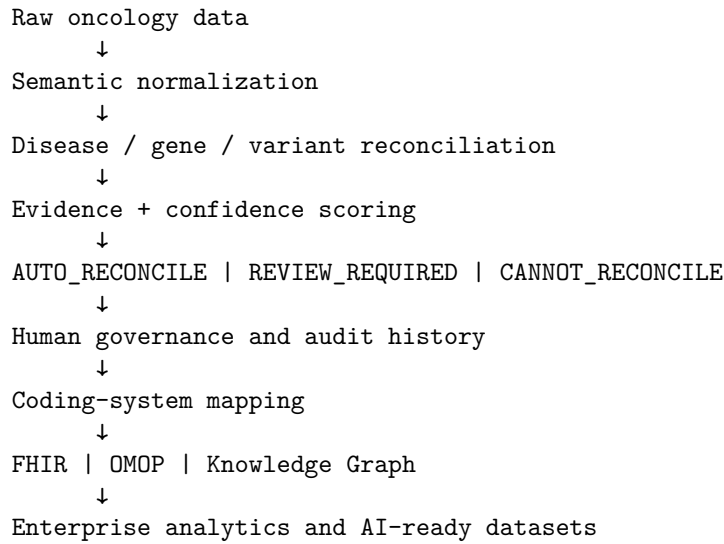
That inconsistency creates expensive manual work and weakens:

- cohort generation;
- biomarker analytics;
- longitudinal patient journeys;
- interoperability;
- real-world-data programs;
- AI training and evaluation.

Trustworthy AI requires trustworthy data.

OncoReconcile AI is a governed data-quality layer that converts fragmented oncology data into standardized, explainable, reviewable, and reusable information.

The Product



The product combines:

- disease, gene, and variant normalization;
- evidence packages and transparent score breakdowns;
- a governed review queue and adjudication history;
- optional LLM suggestions restricted to review-required cases;
- semantic mappings across oncology coding systems;
- synthetic longitudinal patient journeys;
- enterprise, governance, and executive analytics;
- FHIR R4, OMOP CDM v5.4, and knowledge-graph export prototypes;
- interactive APIs and benchmark evaluation.

Working Reconciliation & Evidence

A user submits a disease, gene, and variant record. The platform:

1. normalizes the raw text;
2. checks governed aliases and catalogs;
3. retrieves supporting context and evidence;
4. calculates a confidence score;
5. assigns a safety-aware status;
6. presents evidence, alternatives, notes, and an audit trail;
7. exports the governed result through standards-oriented formats.

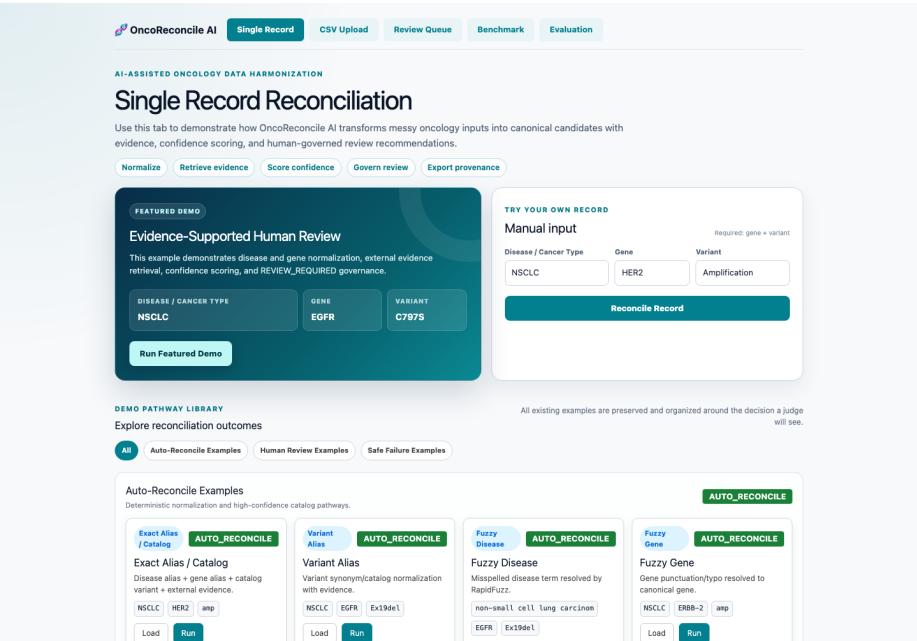


Figure 2: Platform homepage

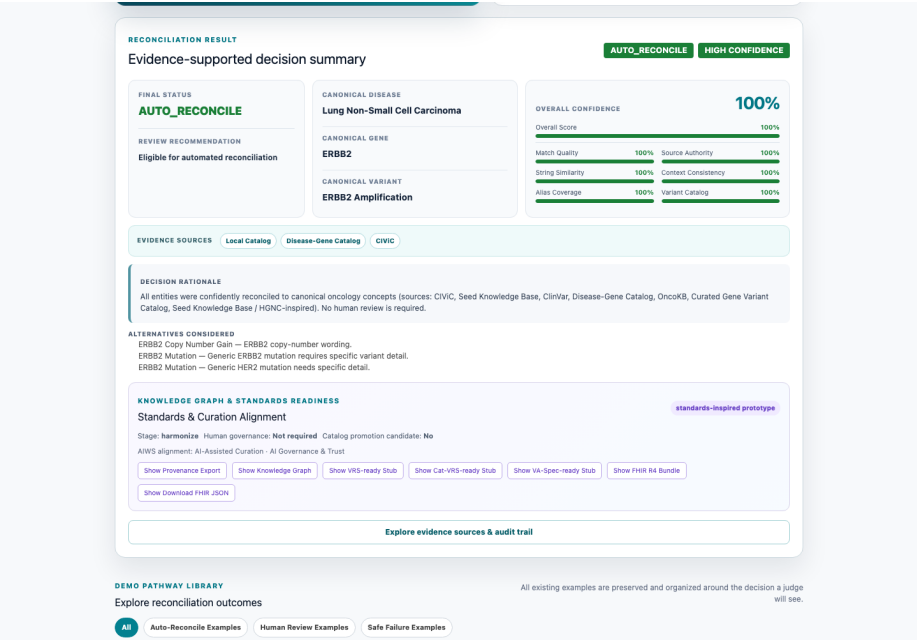


Figure 3: High-confidence reconciliation with evidence

Human-Governed AI

The platform does not force an answer when evidence is incomplete.

Status	Meaning
AUTO_RECONCILE	Evidence supports a high-confidence canonical mapping
REVIEW_REQUIRED	A plausible mapping exists, but a person must decide
CANNOT_RECONCILE	No sufficiently trustworthy mapping is available

Optional LLM assistance is model-agnostic and restricted to review suggestions. It cannot silently promote an uncertain case to automatic acceptance.

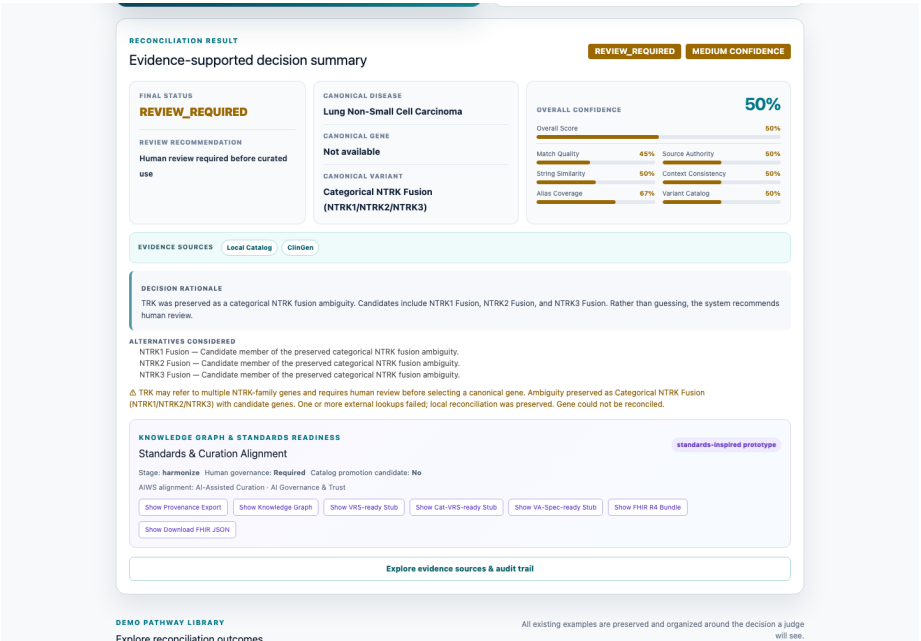


Figure 4: Ambiguous case routed to human review

Enterprise Patient Journey

The enterprise demonstration expands the reconciliation workflow into longitudinal analytics:

- 15 synthetic oncology patients;

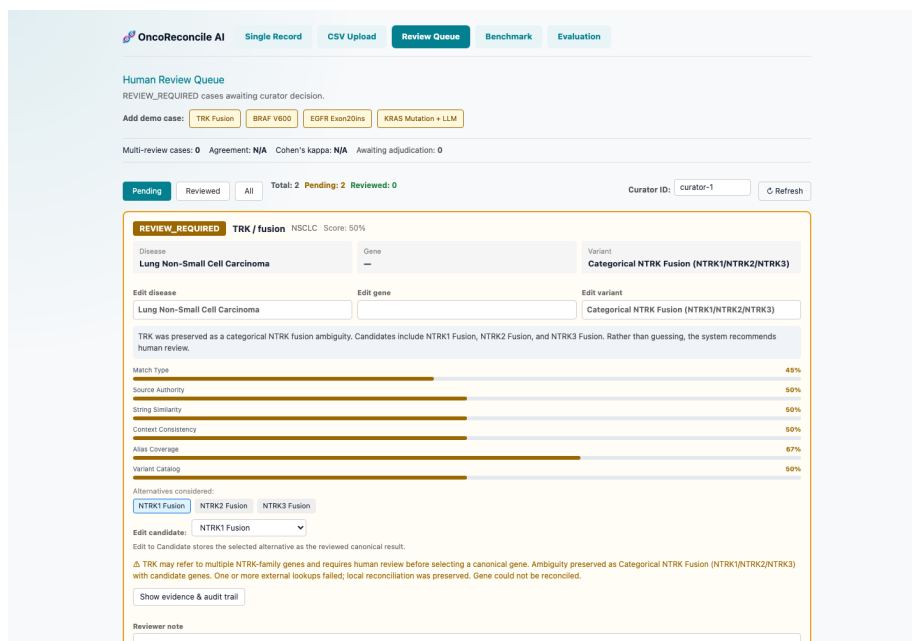


Figure 5: Review queue and curator workflow

- NSCLC, breast cancer, colorectal cancer, melanoma, and AML;
- diagnosis, histology, stage, biomarkers, genomic alterations, treatment lines, response, progression, trial flags, and governance signals;
- patient, cohort, terminology, quality, governance, and executive views;
- expanded FHIR, OMOP, and knowledge-graph exports.

All patient journey data is synthetic and used only for product demonstration.

Semantic Harmonization

Domain	Demonstrated coding systems
Disease	SNOMED CT, NCIt, OncoTree, ICD-10
Histology	SNOMED CT, NCIt
Gene	HGNC
Drug	RxNorm, ATC
Variant / biomarker	ClinVar, ClinGen-inspired identifiers
Laboratory test	LOINC

Mappings retain source, confidence, and review status. These are prototype mappings, not a complete certified terminology service.

Interoperability

The platform demonstrates:

- **FHIR R4 Bundle export:** Patient, Condition, Observation, MedicationStatement, Procedure, and Provenance resources;
- **OMOP CDM v5.4-oriented export:** condition occurrence, measurement, observation, drug exposure, and procedure occurrence records;
- **knowledge graph export:** disease-gene-variant-treatment-evidence relationships;
- **API-first integration:** FastAPI endpoints and interactive OpenAPI documentation.

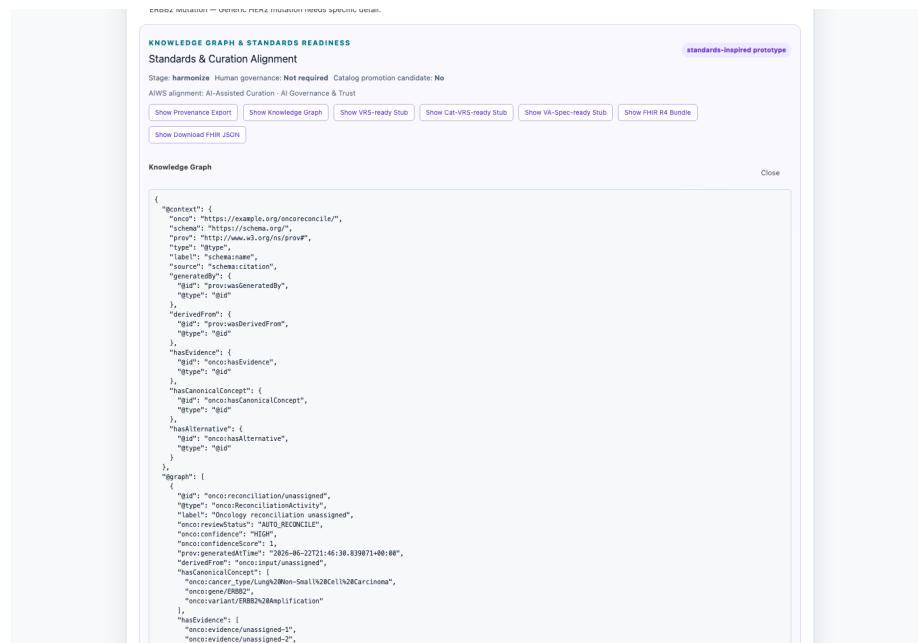


Figure 6: Knowledge graph export

Validation

Validated on June 22, 2026:

Metric	Result
Backend tests	127 passed
Frontend production build	Passing
Benchmark	500 cases
Gene accuracy	96.6%

Metric	Result
Variant accuracy	94.6%
Safety-aware status accuracy	89.8%
False auto-accept rate	0%
Negative-control safety rate	100%

The benchmark is an internal engineering evaluation, not a clinical validation study.

Business Value

OncoReconcile AI targets the costly layer between raw oncology data and useful enterprise data products.

Potential customer value includes:

- less repetitive terminology cleanup;
- faster review of ambiguous records;
- reusable governed mappings;
- stronger traceability and quality reporting;
- easier preparation of analytics- and AI-ready datasets;
- reduced duplication across FHIR, OMOP, research, and warehouse projects.

Target Customers

- cancer centers and health systems;
- molecular diagnostic laboratories;
- pharmaceutical and biotechnology companies;
- clinical research organizations;
- oncology registries and research networks;
- healthcare data and AI platforms.

Commercial Strategy

1. **Paid services and pilots:** data-quality assessment, harmonization, governance design, FHIR/OMOP support.
2. **Team SaaS:** batch workflows, shared review queues, audit dashboards, governed catalogs.
3. **Enterprise APIs:** reconciliation, evidence, governance, exports, and customer-controlled deployment.

The services-first strategy creates early revenue and customer learning before scaling recurring software.



Figure 7: Interactive API documentation

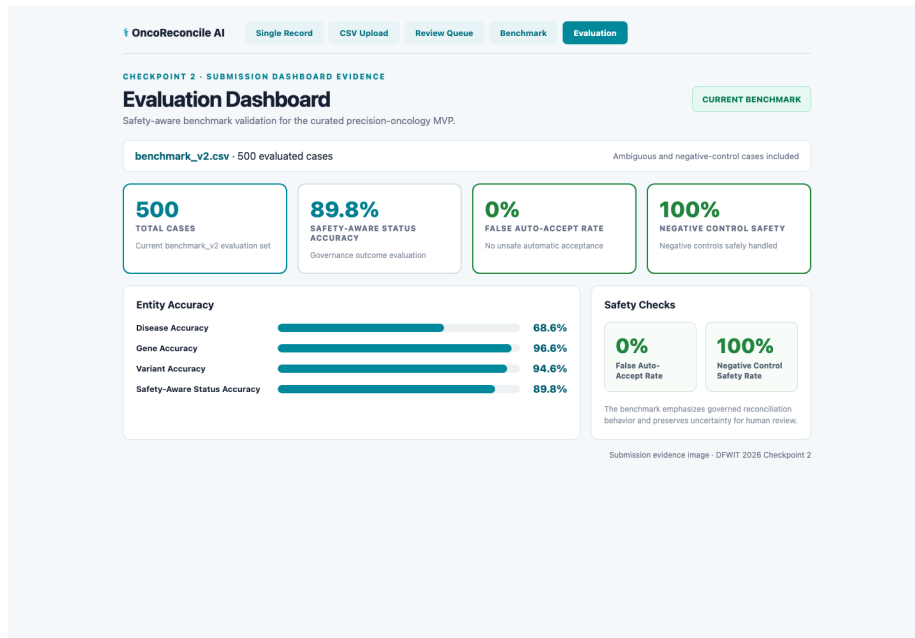


Figure 8: Benchmark and safety evaluation dashboard

Competitive Differentiation

OncoReconcile AI does not attempt to replace authoritative biomedical terminology sources. It operationalizes them through an oncology-specific workflow that combines:

- semantic reconciliation;
- evidence and provenance;
- explicit uncertainty states;
- human governance and adjudication;
- safety-focused benchmarking;
- interoperability exports;
- enterprise patient and executive analytics;
- API-first integration.

Social Impact

Improved oncology data quality can support more reliable research, safer healthcare AI development, more consistent biomarker analysis, and greater reuse of data across organizations.

The product's current impact is data quality and governance. Any downstream clinical benefit requires further validation, clinical oversight, and regulatory assessment.

Team

Variant Vanguard combines healthcare data-engineering insight, software development, product strategy, and a governance-first AI philosophy.

The team's core insight is simple: oncology organizations do not only need a model that produces an answer. They need a data-quality platform that can explain the answer, preserve uncertainty, route difficult cases to experts, and integrate governed results across the enterprise.

Roadmap

- improve disease normalization and governed catalog coverage;
- validate workflow value with de-identified pilot data;
- add authentication, persistent enterprise storage, and role-based access;
- measure curator time, review agreement, and correction rates;
- expand integration and standards-conformance testing;
- scale into a customer-controlled oncology data-quality platform.

Standards Alignment & Reference Resources

OncoReconcile AI is designed to align with widely adopted healthcare interoperability, precision oncology, and biomedical knowledge standards. These resources inform the platform’s semantic harmonization, governance, interoperability, and future expansion roadmap.

Healthcare Interoperability

- HL7 FHIR (Fast Healthcare Interoperability Resources)
- OMOP Common Data Model (OHDSI)
- SNOMED CT
- LOINC
- RxNorm
- ICD-10-CM

Precision Oncology & Genomics Standards

- HGNC Gene Nomenclature
- ClinVar
- ClinGen Allele Registry
- GA4GH Variant Representation Specification (VRS)
- GA4GH Categorical Variation Representation (Cat-VRS)
- GA4GH Variant Annotation Specification (VA-Spec)
- NCI Thesaurus (NCIt)
- OncoTree

Biomedical Knowledge Sources

- CIViC (Clinical Interpretation of Variants in Cancer)
- MyGene.info
- MyVariant.info
- AACR Project GENIE
- ClinGen
- National Center for Biotechnology Information (NCBI)

Platform Positioning

OncoReconcile AI does not replace these standards or knowledge resources. Instead, the platform provides a governance and semantic harmonization layer that helps organizations transform heterogeneous oncology data into explainable, interoperable, and AI-ready assets.

Current implementation includes selected interoperability and terminology concepts. Additional standards integration remains part of the future product roadmap.

Repository Resources

Primary Repository

<https://github.com/oncoreconcile-ai/oncoreconcile-ai>

Submission Branch

<https://github.com/oncoreconcile-ai/oncoreconcile-ai/tree/enterprise-patient-journey-demo>

Key Documentation

- README
- Final Submission
- Architecture
- Architecture Diagrams
- Commercial Strategy
- Roadmap
- Curation Methodology
- Data Curation Pipeline
- Enterprise Patient Journey Demo

Supporting Assets

- Benchmark Evaluation Framework
- Review Queue Governance Workflow
- FHIR Export Prototype
- OMOP Export Prototype
- Knowledge Graph Export Prototype
- Enterprise Analytics Dashboard

Links

Repository:

<https://github.com/oncoreconcile-ai/oncoreconcile-ai/tree/enterprise-patient-journey-demo>

Final submission:

https://github.com/oncoreconcile-ai/oncoreconcile-ai/blob/enterprise-patient-journey-demo/docs/DFWIT_Final_Submission.md

Architecture:

<https://github.com/oncoreconcile-ai/oncoreconcile-ai/blob/enterprise-patient-journey-demo/docs/architecture.md>

Enterprise demo documentation:

https://github.com/oncoreconcile-ai/oncoreconcile-ai/blob/enterprise-patient-journey-demo/docs/enterprise_patient_journey_demo.md

Official DFWIT rules:

<https://ai.dfwit.org/Rules>

Disclaimer

OncoReconcile AI is a prototype biomedical data harmonization, governance, interoperability, and analytics platform. It has not been clinically validated. It is not a medical device, clinical decision support system, diagnostic system, or treatment-recommendation system and does not provide medical advice.

FHIR, OMOP, terminology, and knowledge-graph capabilities are prototypes requiring implementation-specific validation before production use. The enterprise patient journey uses entirely synthetic data; no real patient data is included.

Appendix A – Standards & Reference Resources

OncoReconcile AI is designed to work alongside widely adopted healthcare interoperability, precision oncology, terminology, and biomedical knowledge resources.

Healthcare Interoperability

- HL7 FHIR (Fast Healthcare Interoperability Resources)
- OMOP Common Data Model (OHDSI)
- SNOMED CT
- LOINC
- RxNorm
- ICD-10-CM

Precision Oncology & Genomics Standards

- HGNC Gene Nomenclature
- ClinVar
- ClinGen Allele Registry
- GA4GH Variant Representation Specification (VRS)
- GA4GH Categorical Variation Representation (Cat-VRS)
- GA4GH Variant Annotation Specification (VA-Spec)
- NCI Thesaurus (NCIt)
- OncoTree

Biomedical Knowledge Sources

- CIViC (Clinical Interpretation of Variants in Cancer)
- MyGene.info
- MyVariant.info
- AACR Project GENIE
- ClinGen
- National Center for Biotechnology Information (NCBI)

Enterprise Positioning

OncoReconcile AI does not replace these standards or knowledge resources. It provides the governance, semantic harmonization, evidence, and workflow layer needed to apply them to heterogeneous oncology data and produce explainable, interoperable, and AI-ready assets.

Current implementation demonstrates selected terminology and interoperability concepts. Broader standards integration and conformance testing remain part of the product roadmap.